# Abdelrahman Hanoura

**Data Scientist | AI Engineer | Machine Learning Engineer**

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## Summary

Motivated Data Scientist and Computer Science student with strong analytical, problem-solving, and computational skills. Highly familiar with data analysis, data processing, and algorithm development to create end-to-end ML solutions. Experienced in applying machine learning algorithms, statistics, and optimization methods using Python, Scikit-Learn, and TensorFlow to drive innovation in data science and industrial applications.

## Education

**Tanta University** **Sep 2024 – Jul 2027**
*Bachelor of Science in Computer Science and Artificial Intelligence* *Tanta, Egypt*

- GPA: 3.5 / 4.0 3
- Relevant Coursework: Computer Science, Statistics, Applied Mathematics, Physics, Linear Algebra, Calculus, Algorithms, Scientific Computing.

## Professional Experience

**Digital Egypt Pioneers Initiative (DEPI)** **Jan 2026 – Present**
*Team Leader & AI Intern* *Cairo, Egypt*

- Demonstrated teamwork and leadership by researching solutions to overcome data analytics challenges and guiding technical teams in data science applications.
- Applied algorithm development to create AI workflows using LangChain, LangGraph, and FastAPI, automating processes and meeting with engineers to integrate business rules.
- Developed complex mathematical models that translate technical requirements into functional specifications for industrial applications.

**National Telecommunication Institute (NTI)** **Mar 2026 – May 2026**
*AI Engineer Trainee* *Cairo, Egypt*

- Leveraged strong computational skills to develop computer vision workflows using Python and YOLOv8–v11 for object detection.
- Trained and evaluated machine learning algorithms on a 24,000-image dataset, achieving 94% mAP@50.
- Applied applied mathematics and optimization methods for model development and tuning, improving reported inference latency by 15%.

**National Telecommunication Institute (NTI)** **Aug 2025**
*ML Engineer Trainee* *Cairo, Egypt*

- Designed machine learning pipelines covering data processing, data analysis, feature engineering, and evaluating machine learning algorithms during an intensive 80-hour training.
- Applied statistics, linear algebra, calculus, and mathematical software to overcome data science challenges and develop predictive models achieving 99.90% accuracy.

## Projects

Ayn (أين) – Personal Travel Planner Python, CrewAI, FastAPI, ChromaDB, Tavily API
– Designed and engineered an advanced AI travel platform that curates and richly describes tourist destinations across Egypt, generating dynamic, real-time-updated itineraries.
– Integrated intelligent multi-agent workflows using CrewAI to automatically tailor travel plans based on user-specific constraints, including health conditions and personal preferences.

Customer Churn Prediction Python, XGBoost, Pandas , Random forest

– Developed a customer churn prediction system using XGBoost and machine learning algorithms, achieving 92% accuracy during project evaluation.
– Applied data processing, feature engineering, and optimization methods to improve predictive models.
– Utilized excellent communication abilities to present evaluation metrics, visually and verbally sharing complex data analysis insights with non-technical stakeholders.

Cardiovascular Risk Assessment Model Python, Random Forest, Scikit-Learn

– Developed a machine learning model for cardiovascular risk prediction using 14 clinical variables.
– Applied data processing and statistical modeling techniques utilizing statistics and applied mathematics to prepare the dataset.

Skills

**Data Science & Algorithms:** Machine Learning Algorithms, Algorithm Development, Data Analysis, Data Processing, Optimization Methods
**Core Subjects:** Computer Science, Statistics, Applied Mathematics, Physics
**Programming:** Python, SQL, C++, R
**Machine Learning & Deep Learning:** Scikit-Learn, XGBoost, Random Forest, TensorFlow, Keras, CNNs
**Data & Visualization Tools:** Pandas, NumPy, Matplotlib, Seaborn, Mathematical Software
**Soft Skills:** Strong Analytical Skills, Problem-Solving, Computational Skills, Communication, Teamwork

## Certifications

- Agentic AI — DEPI
- AI for Business — NTI
- AI With Huawei — NTI
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- Python Programming — MaharaTech / ITI Mooca